SCOPING REVIEW TOOL

User Guide

Overview

This guide describes the Scoping Review Tool and its complete six-stage pipeline for importing, screening, characterizing, and analysing academic literature. It explains required inputs, generated outputs, configuration files, and the purpose of each processing stage.

The workflow is organized into six sequential stages, from importing search results to generating the final analyses and visualizations. The navigation menu on the left remains consistent throughout the workflow, while the main panel displays the controls for the selected stage.

Getting Started

The source code, installation instructions, and project documentation are available in the GitHub repository. Installation instructions are provided in the repository's README file.

Quick-Start Options

2stages can be executed without an LLM:

- Stage 1 can be tested by uploading the example datasets provided in the Stage 1/input folder.
- Stage 6 does not require an LLM because it analyses outputs generated in previous stages. The repository includes complete characterization and mapping results for the full corpus, allowing the analysis stage to run immediately.

TIP	Users who want to test the complete LLM pipeline can use the sample input files provided for Stage 1. However, each stage can also be tested independently using either the full dataset or a sample dataset.

File Organization

Each stage has its own input and output folders where intermediate and final files are stored automatically.

Advanced Configuration

Some stages include additional configuration options intended for advanced users. These settings support future extensions of the tool and do not need to be modified for standard use.

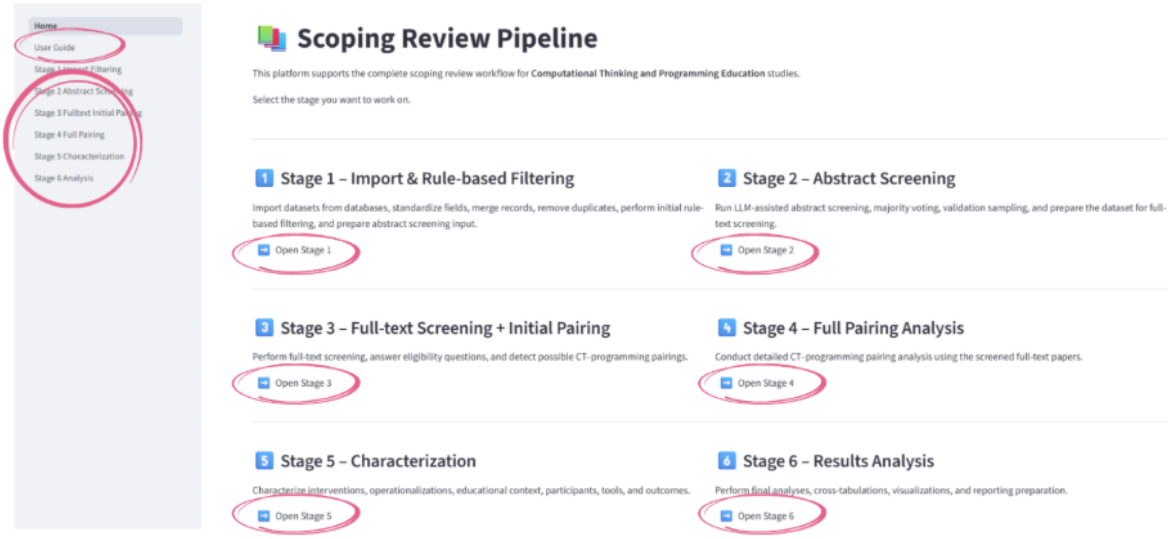

Figure 1. Main interface of the Scoping Review Tool.

User Guide Page

The built-in User Guide page provides an overview of the six-stage workflow, including software requirements, access to required templates and configuration files, expected inputs and outputs for each stage, and general recovery and troubleshooting recommendations. Users are encouraged to review this information before executing the workflow.

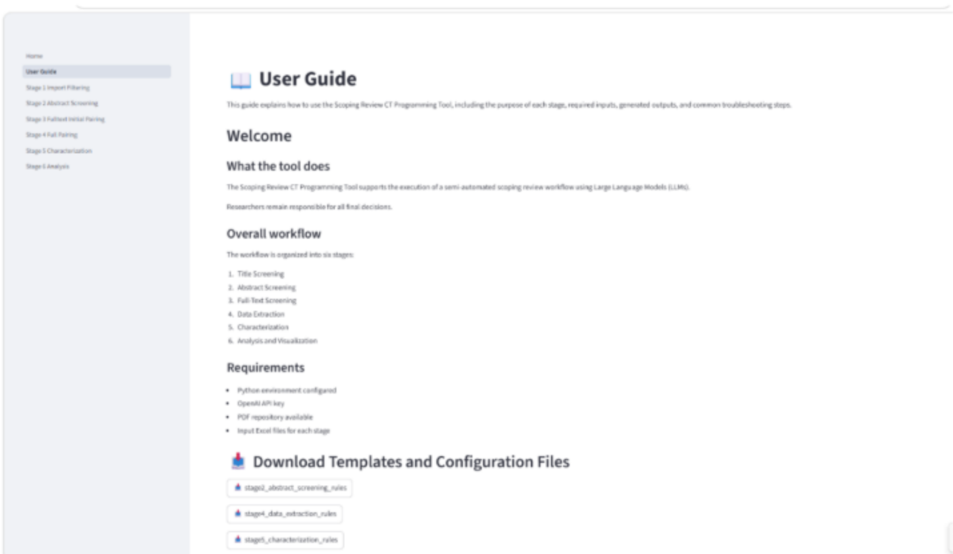

Figure 2. Partial view of the User Guide page

Pipeline Stages

The following sections describe each stage in detail.

Stage 1 | Import, Merge, and Rule-based Filtering

The first stage imports search results exported from the selected digital libraries (ACM, IEEE, ERIC, SCOPUS; ScienceDirect), standardizes the records, removes duplicates, and applies the initial rule-based filtering.

One export file must be uploaded for each selected database. After all datasets have been loaded, click Run Stage 1 to generate the merged dataset that will be used in the next stage.

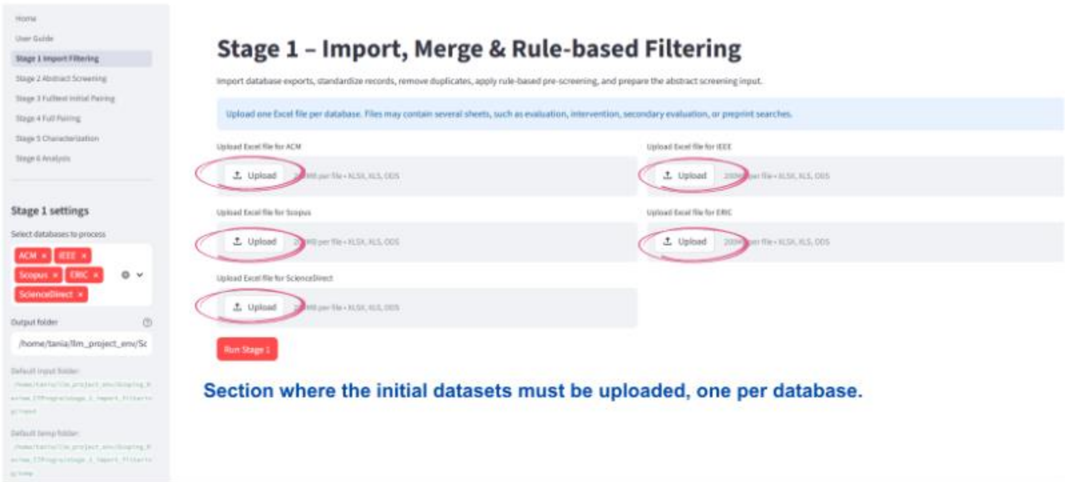

Figure 3. Stage 1 – Title screening page

Stage 2 | Abstract Screening

Stage 2 performs abstract screening using the LLM. Upload the Stage 2 rules template located in the templates folder. By default, the abstract dataset generated in Stage 1 is automatically loaded, although a custom input file can also be selected if required.

Click Run Abstract Screening to start the process.

Pause and Resume

During execution, the screening can be paused after the current paper by selecting Stop after current paper. Clicking Clear stop request resumes the screening.

TIP	Based on our experiments, three iterations were sufficient. The comparison rules applied across three runs minimize the likelihood of overlooking relevant abstracts, though additional iterations increase LLM screening time.

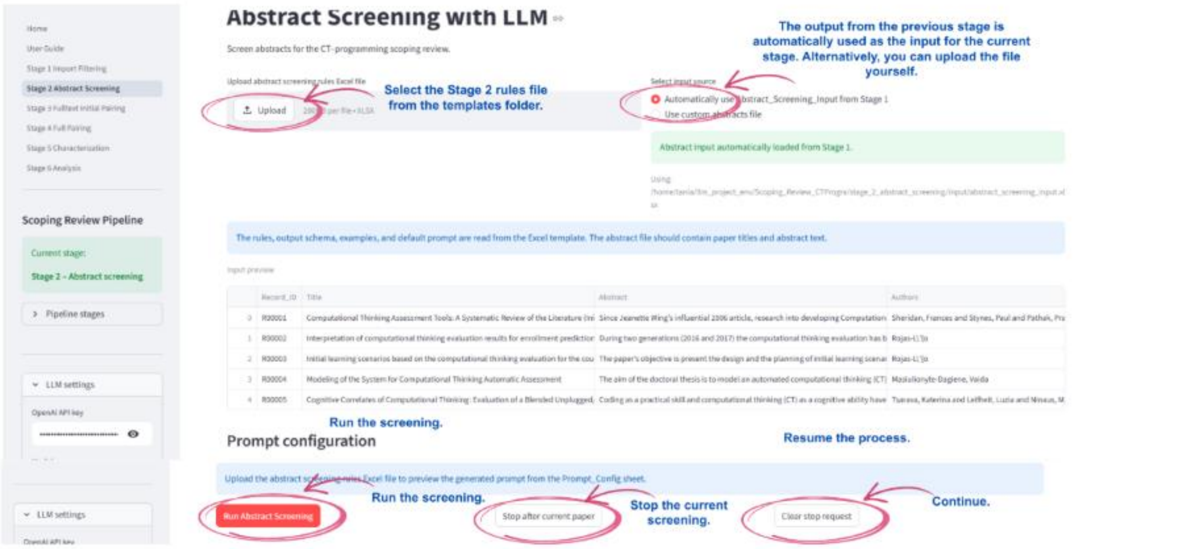

Figure 4. Stage 2 – Abstract screening page

Stage 3	Initial Full-text Screening

Stage 3 performs the first full-text screening. Upload the Stage 3 rules template from the templates folder. The screened papers produced in Stage 2 are automatically used as input, although users may provide a custom dataset if necessary. Once the input has been validated, click Run Full-text Screening to begin.

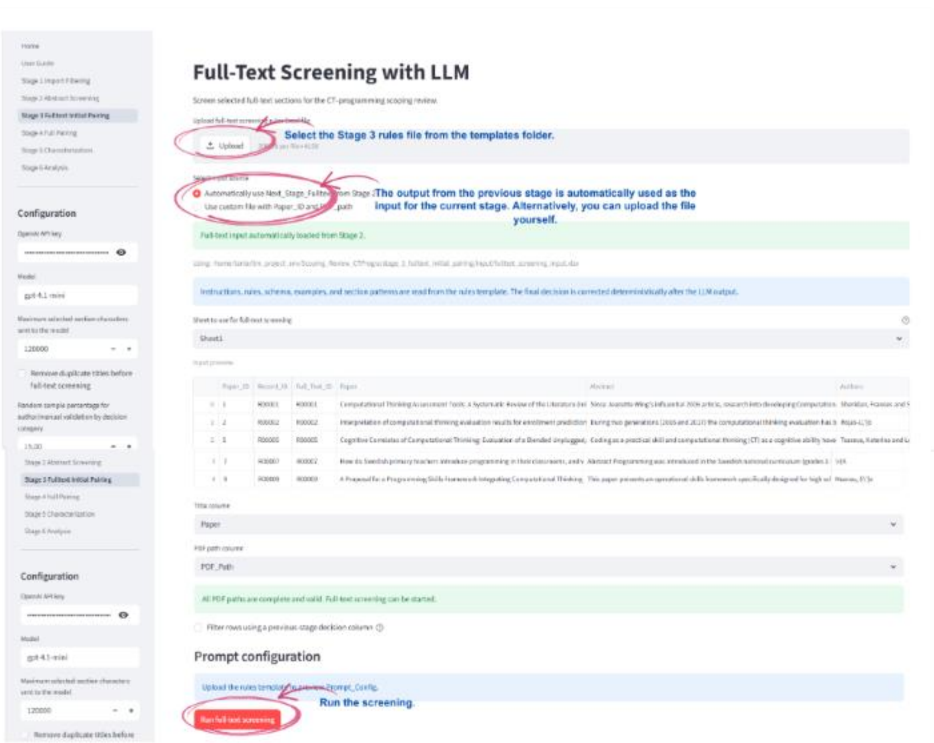

Figure 5. Stage 3 – Full-text screening page

Missing PDFs

If one or more PDFs cannot be found, the interface displays a warning message. Papers without an available PDF are listed so that the user can decide whether to remove them from the screening or retrieve the missing documents manually before continuing.

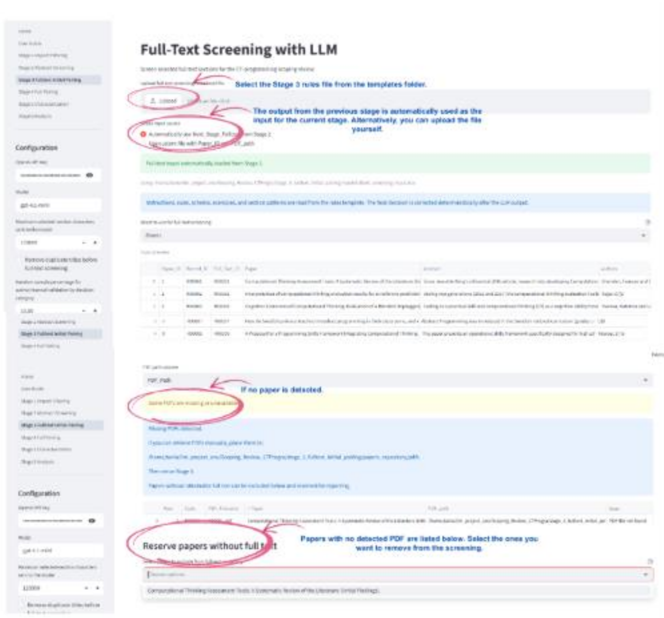

Figure 6. Stage 3 – Full text – Initial pairing screening page / missing pdf

Stage 4	Pairing Process

Stage 4 identifies relationships between Computational Thinking concepts and programming concepts. Upload the Stage 4 rules template from the templates folder. The output generated in Stage 3 is automatically used as input, although a custom dataset may also be selected. Click Run Pipeline to execute the pairing process.

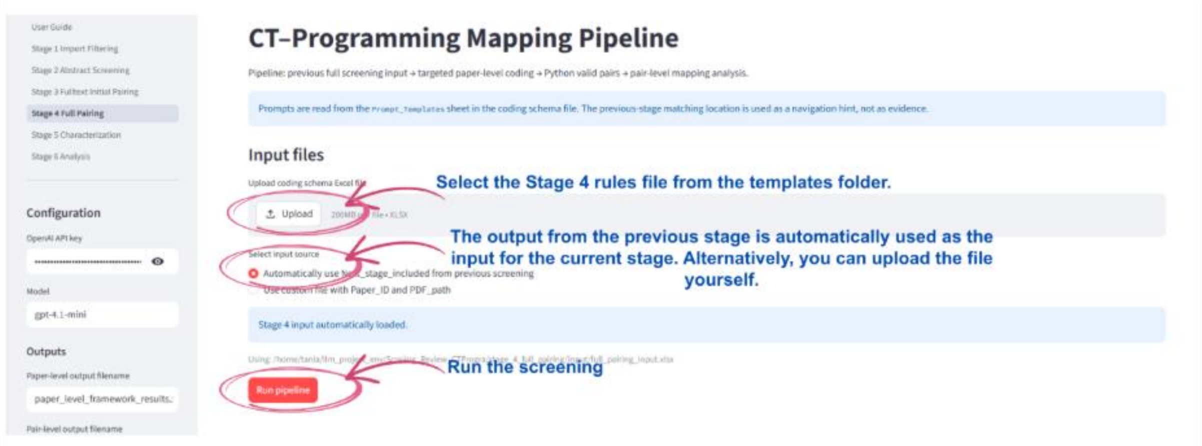

Figure 7. Stage 4 – Pairing page

Stage 5 Paper Characterization

Stage 5 extracts the characterization variables defined in the coding schema. Upload the Stage 5 rules template from the templates folder. The mapping file generated in Stage 4 is loaded automatically. Click Run Characterization to generate the characterization dataset that will be used during the final analysis.

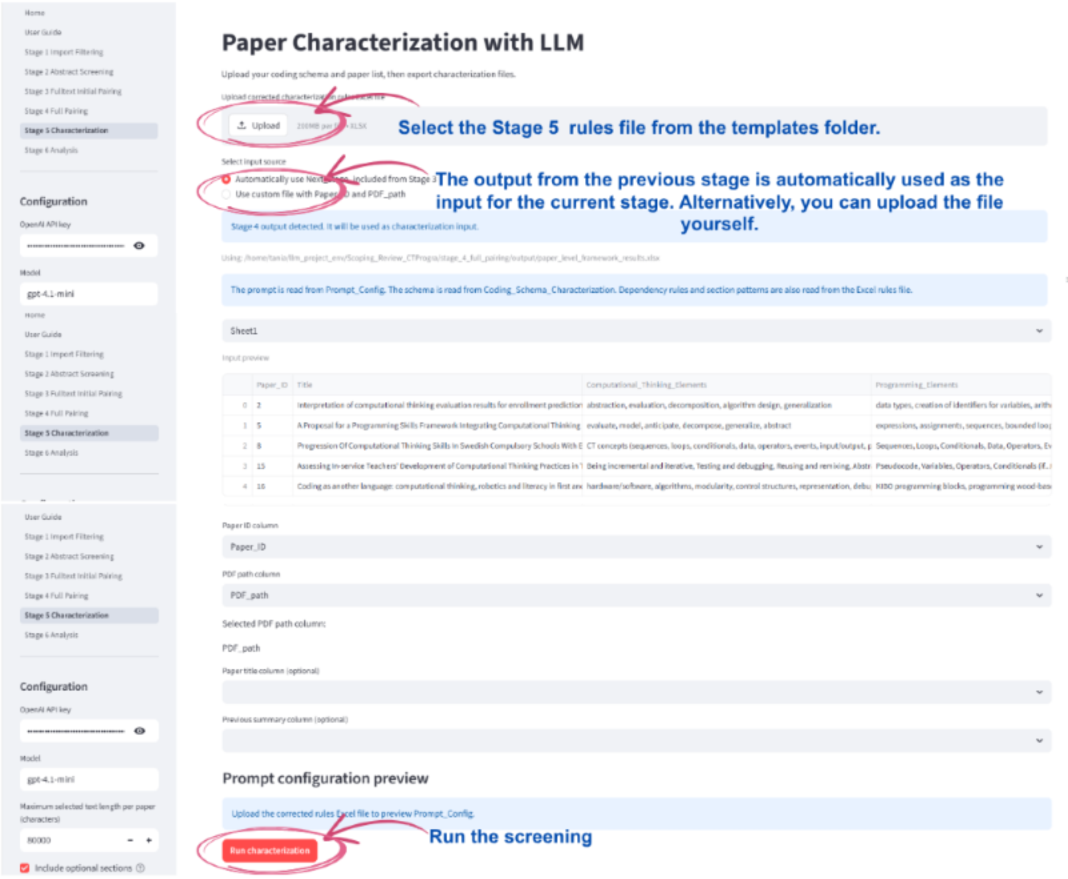

Figure 8. Stage 5 – Characterization page

Stage 6	Results Analysis

The final stage summarizes characterization and mapping results. The characterization dataset, the mapping, the framework-constrained mapping, and the normalization dictionary are automatically loaded from previous stages. Alternatively, users may provide custom input files.

Click Run Analysis to generate all statistical summaries and visualizations. The resulting figures are displayed in the interface and simultaneously saved to their corresponding files in the Stage 6 output folder.

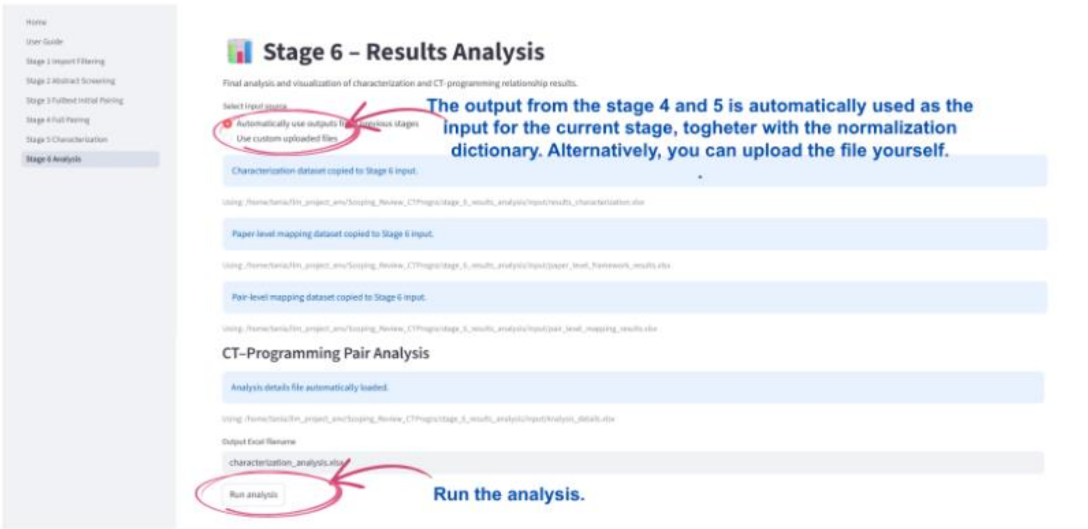

Figure 9. Analysis page form

Stage Summary

Stage	Name	Input	Output
1	Import & Filter	Database exports	Merged dataset
2	Abstract Screening	Stage 1 output + rules	Screened abstracts
3	Full-text Screening	Stage 2 output + rules	Screened papers
4	Pairing	Stage 3 output + rules	Concept mapping
5	Characterization	Stage 4 mapping + rules	Characterization data
6	Results Analysis	Stage 5 data	Statistics & visualizations